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Professional Summary

Applied Scientist at Amazon specializing in context-aware **LLM mid/post-training and rubric-based evaluation** for **agentic systems** — including context extension, grounding, personalization, and multi-turn consistency — on **100B+ MoE** backbones, with experience productionizing **closed-loop post-training data infrastructure** that turns live-traffic failures into continuous model improvements. **Ph.D. in Computer Science** with **20+ publications** (8 first-authored) at **NeurIPS (Spotlight)**, ACL, EMNLP, and NAACL, including **early contributions** to **multi-agent systems** and **set-attention** for tabular language models, one **US patent**, and area chair at **ARR**.

Experience

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| Amazon
<i>Applied Scientist (full-time)</i> | 2024.1 – Present
Bay Area, CA |
| <ul style="list-style-type: none">• Rufus Foundation Modeling (startup-like) — Context-aware Mid/Post-training & Evaluation: Drove the context management and RAG workflow for shopping agents on 100B+ MoE backbones.<ul style="list-style-type: none">– Extended long-context capability from 8K to 128K tokens through continual pre-training (ABF, YaRN), with training-free methods supporting up to 1M tokens, enabling stable context window over extended multi-turn sessions.– Built a verifiable rubric-based eval for personalization helpfulness, raising human alignment from 40% → 78% over 5 iterations; used the eval to drive GRPO optimization, lifting helpfulness from 84% → 93% under user-specific preference settings.– Increased factual accuracy from 67% → 90% on customer-facing RAG evaluations by building synthetic IFT + DPO pipelines for faithfulness and grounding.• Customer Service — Post-training Data Infrastructure & Flywheel: Owned the end-to-end post-training data flywheel for the team's multi-agent LLM system.<ul style="list-style-type: none">– Productionized a scalable annotation and quality control system on Amazon Bedrock for SFT/DPO/GRPO/retrieval/eval data, with sample-level validation (rule-based + LLM-as-judge), dataset-level controls (diversity, de-duplication), and LakeFS-based versioning — adopted across teams as the standard data pipeline.– Developed diversity-aware data selection: agent tool-use accuracy 86% → 93%, training time 4× faster (24h → 6h); introduced difficulty-aware curriculum data selection: response generation quality 88% → 91%, training time 5× faster (10d → 2d).• Intern Mentorship: Mentored two interns on LLM research projects in long-context modeling and multi-turn improvement with verifiable GRPO — 2 papers published, 1 under review. | |
| Amazon Web Services (AWS)
<i>Applied Scientist (Intern)</i> | 2023.5 – 2023.8
Santa Clara, CA |
| <ul style="list-style-type: none">• Developed a collaborative multi-agent prompting framework (CoMM) achieving SOTA on complex science benchmarks in zero-shot and few-shot settings — published at NAACL 2024. | |
| Amazon Web Services (AWS)
<i>Applied Scientist (Intern)</i> | 2022.6 – 2023.1
Santa Clara, CA & remote |
| <ul style="list-style-type: none">• Developed HYTREL, a hypergraph-based tabular LM capturing structural dependencies, achieving SOTA on tabular representation learning — published as NeurIPS 2023 Spotlight (top 5%). | |
| Tencent AI Lab
<i>NLP Researcher (Intern)</i> | 2021.6 – 2021.8
remote (College Station, TX) |
| <ul style="list-style-type: none">• Built a benchmark for zero-shot knowledge base completion across diverse KB sources — published at ICDM 2022 KG Workshop. | |

- Built information extraction and evaluation systems for financial event extraction and causality detection.

Selected Publications

LLM & Foundation Model Training — *Data-centric and structure-aware approaches to LLM training and alignment.*

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. “HYTREL: Hypergraph-enhanced Tabular Data Representation Learning.” *NeurIPS 2023, Spotlight (5%), first work to use set-attention for tabular LMs.*
- Ming Li, **Pei Chen**, Chenguang Wang, et al. “Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning.” *ACL 2025, Findings.*
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. “END: Early Noise Dropping for Efficient and Effective Context Denoising.” pre-print.
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. “ItD: Large Language Models Can Teach Themselves Induction through Deduction.” *ACL-2024.*
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. “Aligning Large Language Models with Implicit Preferences from User-Generated Content.” *ACL 2025.*

Agent — *Multi-agent reasoning, agent-oriented pre-training, and verifiable tool-use training.*

- **Pei Chen**, Shuai Zhang, Boran Han. “CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving.” *NAACL 2024, early work in demonstrating multi-agent systems for complex LLM reasoning.*
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. “Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training.” *NAACL 2025.*
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. “Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates.” *EMNLP 2025.*
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, **Pei Chen**, et al. “Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intents.” accepted to *ACL 2026.*

RAG & Multi-turn & Personalization — *Context-aware conversational LLMs across long-context, retrieval, and personalization modeling.*

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. “LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios.” *NAACL 2025.*
- Ming Li, **Pei Chen**, Zhenhao Zhang, et al. “Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.” Accepted to *ACL 2026.*
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. “ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations.” *NAACL 2025.*
- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. “UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations.” *ACL 2025.*
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. “Leveraging historical information to boost retrieval-augmented generation in conversations.” *Information Processing & Management, March 2026.*
- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. “Instant Personalized Large Language Model Adaptation via Hypernetwork.”, accepted to *ACL 2026.*
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. “POPI: Personalizing LLMs via Optimized Natural Language Preference Inference.” pre-print.

Information Extraction — *Graph-based modeling of non-sequential dependencies in structured extraction (Ph.D. work).*

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition.” *ACL-IJCNLP 2021.*

- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. “Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition.” NAACL 2022.
- **Pei Chen**, Hang Yang, Kang Liu, et al. “Reconstructing Event Regions for Event Extraction via Graph Attention Networks.” AACL-IJCNLP 2020.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Domain-specific Named Entity Recognition via Graph Neural Networks.” US Patent 12,530,529, 2026.

Education

Texas A&M University <i>Ph.D. in Computer Science</i>	2019.8 - 2024.5 College Station, TX
• GPA: 4.0/4.0 • Research Directions: Information extraction and structured NLP (NER, event extraction, knowledge base completion); later extended to LLMs and agentic systems. • Dissertation: Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions. • Teaching assistant: CSCE 636 Deep Learning and CSCE 313 Computer Systems.	
Southwestern University of Finance and Economics <i>Master's Degree in Finance</i>	2016.9 – 2018.6 Chengdu, China
• GPA: 3.9/5.0, ranking 1/178. • Awards: 2017 National Scholarship for Graduate Student.	
National University of Defense Technology <i>B.Engr. in Simulation Engineering</i>	2010.9 – 2014.6 Changsha, China
• GPA: 88.61/100, ranking 1/45. • Thesis: Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.	

Professional Service

- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
- 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
- 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
- 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
- 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
- 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

- **Training Frameworks:** Megatron, Verl, Slime
- **Post-training Methods:** Continual Pre-training/Mid-training (ABF, YaRN), SFT, DPO, GRPO, GSPO, DAPO
- **Evaluation Methodology:** Verifiable rubric design, human alignment calibration (iterative refinement), LLM-as-judge validation, synthetic eval data generation
- **Programming:** Python, PyTorch, SQL, C/C++ , Git, Docker